Learning Illumination for Depth-Resolved Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering fluorescent structures that overlap in projection requires measurements that distinguish their axial arrangements. We learn depth-encoding illumination for fixed-focus three-dimensional fluorescence microscopy. An angle-scanned plane wave passes through a fixed, defocused phase plate, producing depth-dependent excitation while the specimen and detection focus remain stationary. We optimise the plate by minimising weighted squared correlations between detector-integrated voxel responses at different depths and the same lateral position. Training uses the optical forward model without specimen examples or a reconstruction decoder. We evaluate the resulting measurements through comparisons with merged-depth alternatives, controlled crossing filaments, and reconstruction of mouse pyramidal neuron volumes. At eightfold measurement compression and equal deposited excitation dose, the learned plate reduces axial-profile Wasserstein error by 28–50 % across total variation and two plug-and-play methods. In crossing-filament simulations, total variation and pretrained plug-and-play resolve and correctly order structures at a smaller tested separation, 2.56 *rather than* $3.84\,\mu\mathrm{m}$*. The reconstruction gains persist with background and read noise and at reduced dose. These results show that learning the excitation can improve depth separation and volumetric recovery across different reconstruction priors.*

1. Introduction

Computational imaging enables three-dimensional reconstruction from two-dimensional measurements, but recovery depends on how well those measurements distinguish structures at different depths. When different axial arrangements produce similar observations, reconstruction relies more heavily on prior assumptions [2, 6]. In fluorescence microscopy, neuronal branches that overlap in projection may lie at different depths, making this distinction important for recovering their spatial organisation. We investigate how learning the illumination can improve depth discrimination at the measurement stage.

We address this problem by encoding depth in the excitation illumination. A plane wave is scanned in angle through a fixed, defocused phase plate, producing structured illumination patterns that vary with depth (Fig. 2). As the incident angle changes, depth-dependent lateral shifts cause structures at different axial positions to experience different sequences of excitation intensities. A fixed-focus camera records the resulting fluorescence images for computational volume recovery, without axial specimen scanning or detection refocusing.

The effectiveness of this encoding depends on the phase plate. Recent approaches learn optical encoders jointly with reconstruction networks [7, 14, 21], optimising the measurements for a particular decoder. Building on decoder-independent optical design [17, 20, 24], we optimise the plate through the optical forward model. Our objective reduces weighted squared correlations between detector-integrated measurement signatures of voxels at different depths and the same lateral position. Each signature includes the excitation, emission blur and pixel averaging across illumination angles. Training requires neither specimen examples nor a reconstruction solver.

We assess whether this optical design improves the recovery of axial arrangements. A correct intensity centroid alone is insufficient: two separated layers and one merged layer can share the same mean depth. We therefore evaluate the complete axial profiles, alongside centroid error and image quality. Measurement-level comparisons test the separation of neuronal volumes from merged-depth alternatives before reconstruction. Controlled crossing filaments test separation and axial ordering (Fig. 1), while simulated measurements of mouse pyramidal neuron volumes test recovery with total variation and plug-and-play methods using pretrained or neuron-trained denoisers. We also examine measurement and dose trade-offs, alternative optical objectives, noise and alignment.

Our contributions are threefold:

1. **Learning depth-discriminative illumination.** We develop an angle-scanned excitation encoder with a fixed, defocused phase plate, and learn its phase using detector-integrated depth signatures without specimen examples or a reconstruction decoder.
2. **Evaluating axial separation.** We connect measurement-

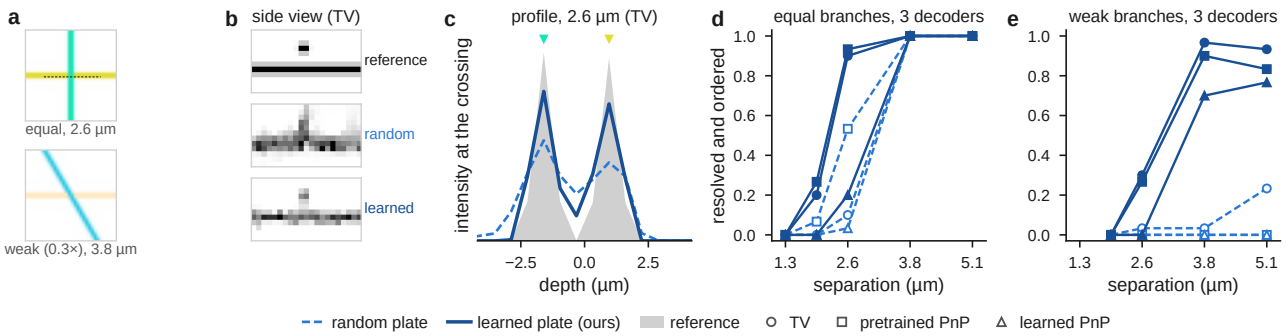


Figure 1. **Learned illumination improves separation of structures that overlap in projection.** (a) Crossing-filament references. (b) An axial section reconstructed with TV, independently scaled for visibility, and (c) crossing profiles averaged over three geometries. (d, e) Separation and correct-ordering rates for equal and unequal branch intensities, pooled across geometries, axial centres and measurement draws. At equal dose and eightfold measurement compression, the learned plate reduces the tested separation threshold under TV and pretrained PnP; learned PnP retains the same threshold.

level comparisons with controlled reconstruction tests, showing increased separation from merged-depth alternatives and separation and correct ordering of crossing filaments at smaller tested gaps under two reconstruction methods.

3. **Validating volumetric recovery across priors.** We demonstrate improved axial-profile recovery with TV and two PnP configurations, and assess the benefits under fewer measurements, reduced excitation dose and optical perturbations.

2. Related work

Depth encoding in fluorescence microscopy. Three-dimensional fluorescence microscopy can encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D recover volumes from depth-dependent detection responses [1, 4, 24]. Excitation-based methods instead use illumination patterns that vary with depth. Yeh *et al*. [25] combine translated speckle patterns with axial scanning, while Pascucci *et al*. [15] exploit axial speckle decorrelation and fluorescence saturation for volumetric recovery from a two-dimensional scan. Related approaches use angular scanning through a fixed scattering layer for two-dimensional fluorescence recovery [26], or tilted pattern projection to produce depth-dependent lateral shifts [8]. We study linear fluorescence with angular scanning through a fixed, defocused phase plate, and learn its phase profile to improve axial discrimination.

Learning optical encoders. Optical design can be guided by reconstruction performance or by properties of the measurements. End-to-end methods jointly optimise optics and reconstruction networks [7, 14, 21], while Kellman *et al*. [10] optimise illumination for a specified reconstruction algorithm. Decoder-independent methods instead optimise optical responses directly. Shechtman *et al*. [20] use Fisher information for 3D localisation, while Miniscope3D [24] optimises a detection-side phase mask by reducing cross-coherence between depth-dependent point-spread functions. Similar coherence objectives appear in sensing-matrix design [5]. IDEAL and IDEAL-IO [13, 17] optimise estimated mutual information using fitted measurement-distribution models, without a reconstruction decoder. Our excitation-side objective follows this measurement-driven principle. It compares detector-integrated voxel responses across illumination angles, reducing weighted squared correlations between depths at each lateral position. Training needs no specimen examples, fitted measurement distribution or reconstruction solver. We evaluate the resulting design through axial-profile recovery and controlled separation tests.

Volumetric reconstruction and priors. Recovering three-dimensional structure from limited measurements requires assumptions about the specimen. Total variation promotes piecewise-smooth structure [19], while plug-and-play (PnP) methods incorporate denoisers as implicit priors [9, 22]. Multi-slice fusion combines lower-dimensional denoisers acting along complementary orientations of a shared volume [11]. We adapt this framework to fluorescence reconstruction, combining measurement consistency, slice-wise denoising and sparsity regularisation. Stronger priors can improve recovery, but may favour plausible structures that the measurements cannot distinguish [2, 6]. We therefore evaluate the same learned optical encoder with conventional regularisation and learned priors, and test measurement distinguishability before reconstruction. Axial-profile and crossing tests assess the recovery of depth arrangements alongside conventional image-quality metrics.

3. Learning depth-encoding illumination

Acquisition. We vary the illumination direction across two angular coordinates while keeping the phase plate fixed (Fig. 2). For transverse position $\mathbf{r} = (x, y)$, the incident plane wave is $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^\top \mathbf{r}}$, with $\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t)$, where α_t is the azimuth and θ_t parametrises the direction in the vertical plane containing that azimuth. The corresponding forward direction is $(\mathbf{s}_t, \sin\theta_t)$, and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for the simulated refractive index of one. Thus, the two coordinates control the orientation and signed magnitude of the transverse phase gradient. The scan contains 256 distinct directions with $0.5 \le \|\mathbf{s}_t\| \le \sqrt{3}/2$. Each incident field passes through the phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The NA-0.8 objective pupil filters the modulated field before a vectorial Richards–Wolf relay [12, 18] produces the specimen excitation $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$, assuming x-polarised illumination. Coordinates and distances are expressed in specimen-equivalent units. The camera records one fluorescence image per direction at a fixed detection focus. Acquisition is sequential, not single-shot.

Why the angle encodes depth. Propagation after the phase plate produces a spatially structured excitation pattern. Changing the illumination angle shifts this pattern laterally, with displacement increasing with propagation distance. In the paraxial thin-screen approximation, a change $\Delta\mathbf{s}$ in transverse direction cosine produces a displacement $\Delta\mathbf{r} \approx d\,\Delta\mathbf{s}$ after propagation over distance d. Different depths therefore experience different sequences of excitation intensities during the angular scan. Plate defocus places the shear pivot outside the specimen. This model explains the encoding principle, while our high-NA vectorial model accounts for finite-pupil effects and departures from ideal translation. Section 6.1 evaluates the resulting shear.

Image formation. Let $X_0(\mathbf{r}, z)$ denote the specimen's fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$ [25]. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z') X_0(\mathbf{r}', z') I_t(\mathbf{r}', z')\,\mathrm{d}^2\mathbf{r}'\,\mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity point-spread function and Ω is the specimen volume. Pixel p, occupying region D_p, records the normalised sample $\hat{Y}^0_{t,p} = |D_p|^{-1}\int_{D_p} g_t(\mathbf{r})\,\mathrm{d}^2\mathbf{r}$. Fluorescence contributions therefore add incoherently before pixel averaging. Multiplication by the excitation pattern mixes specimen spatial frequencies before detection blur, so the camera need not resolve the illumination pattern itself. Depth discrimination depends on how these measurements vary across illumination angles.

Voxel-integrated operator. To capture excitation variations within reconstruction voxels, we evaluate the optics on a finer grid and integrate the measurement responses over each voxel. Let B be the fine-grid measurement operator, including emission propagation and pixel averaging, and let U replicate each reconstruction coefficient over its fine-grid cells. Then $\mathsf{A} = \mathsf{B}\mathsf{U}$ preserves the fine-grid response for every specimen represented in this piecewise-constant basis.

Sub-voxel specimen structure remains unrepresented. For a fine-grid specimen $\mathbf{x}_f$, its coarse representation $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, obtained by voxel averaging, satisfies

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{U}\mathsf{R}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{2}$$

Thus, integration retains within-voxel optical variation, while $\boldsymbol{\delta}_{\mathrm{disc}}$ captures specimen approximation error.

Linear, incoherent fluorescence enables efficient assembly:

$$\mathsf{A}_{(t,p),j} = \sum_{i\in\mathcal{P}_j} I_t(\mathbf{r}_i, z_i)\, g_{p,i}, \tag{3}$$

where $\mathcal{P}_j$ contains voxel j's fine-grid cells, $(\mathbf{r}_i, z_i)$ is the location of cell i, and $g_{p,i}$ is its emission footprint at pixel p, including its integration weight. These footprints are computed once and reused across angles. After precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF\log F)$ for constructing columns through separate FFT-based forward simulations, with T patterns, P detector samples, F fine-grid cells and N reconstruction voxels.

Learning the plate. We optimise the plate phase ϕ to reduce correlations between measurement signatures at different depths, with the aim of improving depth discrimination. For a voxel at lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ denote its centred, unit-norm operator column (Eq. (3)), collecting all camera samples across illumination angles. The signature therefore includes excitation modulation, emission blur and pixel averaging. We minimise

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r}} \sum_{z\neq z'} w_{zz'} \left(\mathbf{a}_{\mathbf{r},z}^\top \mathbf{a}_{\mathbf{r},z'}\right)^2, \tag{4}$$

where valid pairs mirrored about the detection focus receive weight two and other pairs receive weight one. Squaring penalises correlations of either sign; normalisation prevents uniform amplification of the signatures from reducing the loss.

Training differentiates through the excitation, vectorial relay and detection model, without specimen examples or a reconstruction solver. The objective is a surrogate for axial discrimination. It does not directly optimise filament

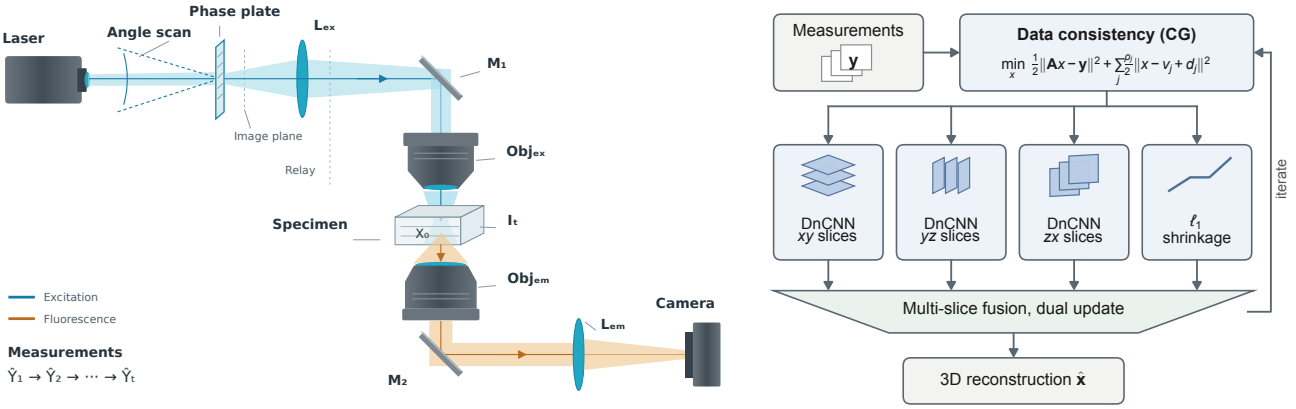


Figure 2. **Fixed-focus acquisition and reconstruction.** Left: an angle-scanned plane wave illuminates a fixed, defocused phase plate; a high-NA relay maps the field into the specimen, and a fixed-focus camera records one fluorescence image per angle. Plate distances are in specimen-equivalent units. Blue: excitation; orange: fluorescence. Right: multi-slice PnP combines a data-consistency update, slice-wise denoising and sparsity on a shared volume.

separation or guarantee stable recovery of a volume; these are evaluated after training, with the plate fixed. Training details are given in the supplement.

4. Reconstruction methods

We evaluate the learned optical encoder using three reconstruction methods: total variation (TV), plug-and-play (PnP) with a pretrained denoiser, and PnP with neuron-trained denoisers (learned PnP). None is trained jointly with the optical encoder. Each method uses the corresponding forward operator, while its prior remains fixed across encoders. Scalar reconstruction parameters are selected separately for each encoder and compression setting using held-out validation volumes.

Total variation. As a non-learned baseline, we reconstruct the volume using box-constrained least squares with isotropic three-dimensional total variation regularisation [19]. The resulting optimisation problem is solved using the alternating direction method of multipliers (ADMM) [3].

Multi-slice plug-and-play. PnP reconstruction combines the physical measurement model with denoisers acting as implicit priors [9, 22]. Following multi-slice fusion [11], we denoise a shared three-dimensional volume along complementary orientations. For $o \in \{xy, yz, zx\}$, P_o rearranges the volume into two-dimensional slices, which $\mathcal{D}$ processes independently, giving $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$.

We combine measurement consistency, denoising and sparsity through PnP-ADMM. Let $\mathcal{J}$ index the active branches, with penalties ρ_j, auxiliary volumes $\mathbf{v}_j$ and scaled dual variables $\mathbf{d}_j$. Each iteration solves

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \textstyle\sum_{j\in\mathcal{J}} \frac{\rho_j}{2}\|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2 \tag{5}$$

using conjugate gradients. The branches then apply denoising or soft thresholding, followed by

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}.$$

Pretrained PnP uses a natural-image DnCNN [27] across the three slice orientations and a sparsity branch. *Learned PnP* replaces this denoiser with a neuron-trained 2D network shared across orientations and adds a volumetric branch with a neuron-trained 3D network [9, 28]. Both variants retain the same measurement-consistency formulation. We initialise from normalised back-projection and clip the final estimate to $[0, 1]$. Iteration counts, penalties and parameter-selection procedures are specified in the supplement.

5. Experimental setup

All experiments use the simulated fixed-focus fluorescence microscope described in Sec. 3. Excitation and emission wavelengths are set to 532 nm, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has 0.32 μm features and is positioned 6.6 μm before the intermediate illumination image plane, in specimen-equivalent coordinates. We use 256 illumination directions. Optical fields are evaluated with 0.08 μm lateral and 0.32 μm axial sampling, followed by 0.32 μm analysis sampling and 0.64 μm reconstruction voxels. The model assumes linear fluorescence in a homogeneous, non-scattering medium.

Volumes and compression. We reconstruct volumes on a $16 \times 48 \times 48$ voxel grid ($N = 36{,}864$), covering 30.7 $\times$

$30.7 \times 10.2\,\mu\mathrm{m}^3$ (lateral × lateral × axial). Each camera image contains 12×12 samples. Using $k = 256$, 64, or 32 illumination directions gives measurement compression factors $C = N/(144k)$ of 1, 4, and 8, respectively. For $C > 1$, we evaluate five randomly selected direction subsets. Reference volumes are obtained from two-photon stacks of mouse pyramidal neurons in the public DEEP^2 dataset [23], min–max normalised and resampled to the analysis grid.

Dataset splits and validation. The test set contains six neuronal cells from three animals, with two prespecified crops per cell selected by a structure score. With reconstruction parameters fixed, we randomly selected eight additional crops per cell, giving 60 test crops for the cross-decoder evaluation (Sec. 6.4). Unless otherwise stated, other full-field experiments use the 12 prespecified crops. Neuron-trained denoisers are trained on 459 crops from a fourth animal. Reconstruction parameters are selected separately for each encoder and compression setting using two vascular crops from a separate stack acquired from a different animal. No test volumes are used for training, tuning, or calibration.

Encoders and photon budget. We compare three random phase-plate realisations, a defocused binary amplitude mask at the plate plane, angular scanning without a plate, and an in-focus DMD with random binary patterns. All encoders share the same relay, pupil, detection model and forward-operator assembly. Excitation is simulated over a field larger than the specimen to avoid beam-edge effects and normalised to a common mean over the full angular scan and specimen volume. For k illumination directions, photon counts follow $\mathrm{Poisson}(\kappa_0\, 256\, \bar{y}_i/k)$, where $\bar{y}_i$ is the noiseless measurement. The factor $256/k$ increases exposure per image as the number of directions decreases, maintaining the nominal total deposited dose. For each specimen, κ_0 is calibrated using random realisation 0 to give 1000 expected photons per pixel per image at the mean signal for $k = 256$, and is shared across encoders. The learned plate is initialised from this realisation, optimised within the specimen field, and trained with two optimisation seeds that differ in their sampled positions and directions.

Axial-profile and image-quality metrics. Our primary depth metric is the axial Wasserstein-1 distance [16]. At each lateral column c, we normalise the non-negative reference and reconstructed axial profiles and form their cumulative distributions F_c and $\hat{F}_c$. The reported error is

$$E_{\mathrm{W1}} = \frac{\Delta z}{\sum_{c\in\mathcal{S}} w_c} \sum_{c\in\mathcal{S}} w_c \sum_{\ell} \left| F_c(z_\ell) - \hat{F}_c(z_\ell) \right|, \qquad (6)$$

where $\Delta z = 0.64\,\mu\mathrm{m}$, w_c is the reference column brightness, and $\mathcal{S}$ contains columns exceeding 10 % of the maximum reference column brightness. This measures axial-profile displacement in micrometres and detects merging that can preserve the centroid. We also report the signal-weighted absolute centroid error over the same columns, PSNR and foreground SSIM. W1 normalises column brightness; the image-quality metrics retain sensitivity to intensity and lateral errors. Two depth-blind references retain the lateral projection while assigning every column either the crop-averaged axial profile or a uniform depth profile.

Measurement distinguishability. We compare each reference volume x with a merged alternative x' that preserves its lateral projection and column centroids but collapses the layered columns in depth. Measurements are generated using the fine-grid model at equal deposited dose. For the shot-noise model, we compute

$$\mathcal{K}_\phi(x, x') = \min_{\beta>0} D_{\mathrm{KL}}\big(\mathrm{Pois}(\boldsymbol{\mu}_\phi(x)) \,\|\, \mathrm{Pois}(\beta\boldsymbol{\mu}_\phi(x'))\big)\,, \qquad (7)$$

where $\boldsymbol{\mu}_\phi$ denotes expected photon counts across all images. Fitting β removes the alternative's unknown global brightness. This comparison uses no reconstruction prior. It tests the selected alternative, rather than all possible volumes. We evaluate the 59 neuronal crops containing layered columns, with additional phantom and background-noise comparisons in the supplement.

Crossing structures. Two Gaussian filaments cross in projection at different depths. We test three crossing geometries, two axial centres, and five measurement draws at $C = 8$, using reconstruction parameters selected on the validation data. Success requires resolving both depths at the crossing and correctly ordering the filament arms. The separation threshold is the smallest tested gap with at least 80 % success at each axial centre and every larger tested gap. A merged-filament control checks false splitting. Unequal-brightness and three-filament cases test more difficult configurations; detailed peak and ordering rules are given in the supplement.

Paired comparisons and statistics. Encoders are paired on specimens, illumination-direction subsets, excitation dose and noise protocol. We average measurement draws within each crop before comparing encoders. Confidence intervals use 10,000 bootstrap resamples of the six cells, retaining their crops together. Relative error reductions compare the paired means; KL ratios are summarised geometrically. All profile-table metrics are computed from the same reconstructions. Re-evaluation checks and the additional reduced-field and uncertainty protocols are detailed in the supplement.

6. Results

6.1. Depth encoding before learning

We first examine the untrained encoding mechanism, comparing the random phase plate with a defocused amplitude mask, a plate-free angular scan and an in-focus DMD. The phase plate produces approximately linear depth-dependent lateral shifts, with a shear pivot near its conjugate plane. Excitation sequences at fixed lateral positions decorrelate over approximately $1\,\mu\mathrm{m}$ of axial separation. Detailed excitation and point-emitter diagnostics are given in the supplement.

With 32 illumination directions, maximum-likelihood localisation of point emitters with unknown brightness gives depth misassignment rates of 1.5–1.8 % for the phase plates, compared with 4.4–5.3 % for amplitude masks, 25 % for the tested DMD and 89 % without a plate. In neuronal reconstruction, the amplitude mask is competitive in centroid depth error, although the random phase plate achieves 0.4–1.2 dB higher PSNR at $C = 8$. These controls establish a useful untrained encoder. The following comparisons isolate the effect of learning by pairing the learned plate with its own random initialisation.

6.2. Measurements distinguish axial arrangements

The phase maps and measurement signatures change during training (Fig. 3). Mean absolute depth correlation falls between detector-integrated depth signatures from 0.84 to 0.48 at 32 illumination directions. To test a consequence beyond the training statistic, we compare each neuronal volume with an alternative that merges its layered columns while preserving their brightness and centroids (Fig. 3). The learned plate increases brightness-profiled Poisson KL by a geometric mean factor of 10.0 (95 % CI: 8.7–11.8), with an increase in all 59 eligible crops. The corresponding factor is 10.4 with added background.

The gain is present before reconstruction. It measures stronger separation from these axial alternatives, without establishing unique recovery or showing that a merged volume agrees with the random plate's measurements. The phantom comparisons show the same trend, although large KL alone does not ensure recovery at every gap.

6.3. Resolving crossing structures

The crossing-filament experiment tests whether improved measurements translate into recovery of distinct structures (Fig. 1). With TV and pretrained PnP, the smallest tested gap meeting the resolution and correct-ordering criterion falls from 3.84 to $2.56\,\mu\mathrm{m}$. The improvement holds at both axial centres, including the centre that does not coincide with the mirror pairing emphasised during training. Neither plate falsely splits the merged control in the tested cases.

Learned PnP retains a $3.84\,\mu\mathrm{m}$ threshold, although learning increases success rates at smaller gaps. Learning also helps recover the weaker branch in unequal-intensity crossings, while closely spaced three-filament cases remain difficult. Thus, the learned plate improves controlled separation under two methods; it does not provide a common resolution threshold across all priors.

	axial W1 (μm) ↓		reduction [95 % CI]
decoder	random	learned	learned vs random
TV	0.371	0.218	41 % [38, 45]
pretrained PnP	0.579	0.290	50 % [47, 53]
learned PnP	0.271	0.195	28 % [26, 29]

Table 1. **Axial-profile recovery at eightfold measurement compression.** Mean over 60 crops from six cells, with five direction subsets per crop and equal deposited excitation dose. Relative reductions compare paired means; intervals resample cells. All three methods improve on every crop.

6.4. Axial-profile recovery across reconstruction priors

The learned plate improves mean axial W1 under TV, pretrained PnP and learned PnP by 41 %, 50 % and 28 %, respectively (Fig. 4 and Tab. 1). All cell-bootstrap intervals exclude zero, and all 60 crops improve under each method. The evaluation includes 48 randomly selected crops added after decoder parameters were fixed, alongside the 12 prespecified crops. A second optical optimisation seed gives comparable W1 reductions of 41 %, 50 % and 29 % from the same initial plate.

Centroid error also falls by 30–54 %, and PSNR increases by 1.6–1.8 dB. Both the axial distributions and their centroids are recovered more accurately. On layered columns, the strictly reproduced TV comparison gives a 34 % W1 reduction; detailed subset results and re-evaluation checks are provided in the supplement.

The in-focus DMD remains a useful control for weak axial encoding. On the 12 prespecified crops, its W1 is 2.24–$2.38\,\mu\mathrm{m}$ across the three decoders, close to the uniform-depth reference of $2.27\,\mu\mathrm{m}$. These results concern the tested in-focus encoder, rather than DMD-based imaging systems in general.

6.5. Measurement and dose trade-offs

On the 12 prespecified crops, the learned plate lowers W1 at all tested compression factors ($C = 1, 4, 8$). At equal total deposited dose, its 32-direction measurements outperform the random plate's 256-direction measurements under TV (0.196 versus $0.249\,\mu\mathrm{m}$) and pretrained PnP (0.277 versus $0.413\,\mu\mathrm{m}$). Learned PnP gives similar means (0.206 versus $0.203\,\mu\mathrm{m}$), without evidence establishing equivalence. Here, fewer images receive longer exposures, so the measurement

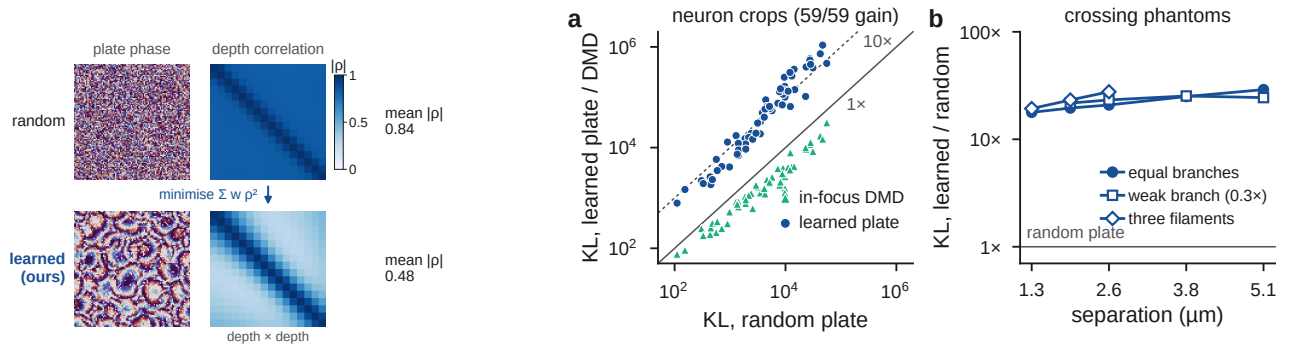


Figure 3. **Learning changes the plate and increases measurement separation.** Left: phase maps and absolute depth-signature correlations before and after training. Right: brightness-profiled Poisson KL for (a) neuronal crops and (b) crossing phantoms, comparing each reference with a merged-depth alternative. Crop values average the five direction subsets. These tests use the expected measurements before reconstruction.

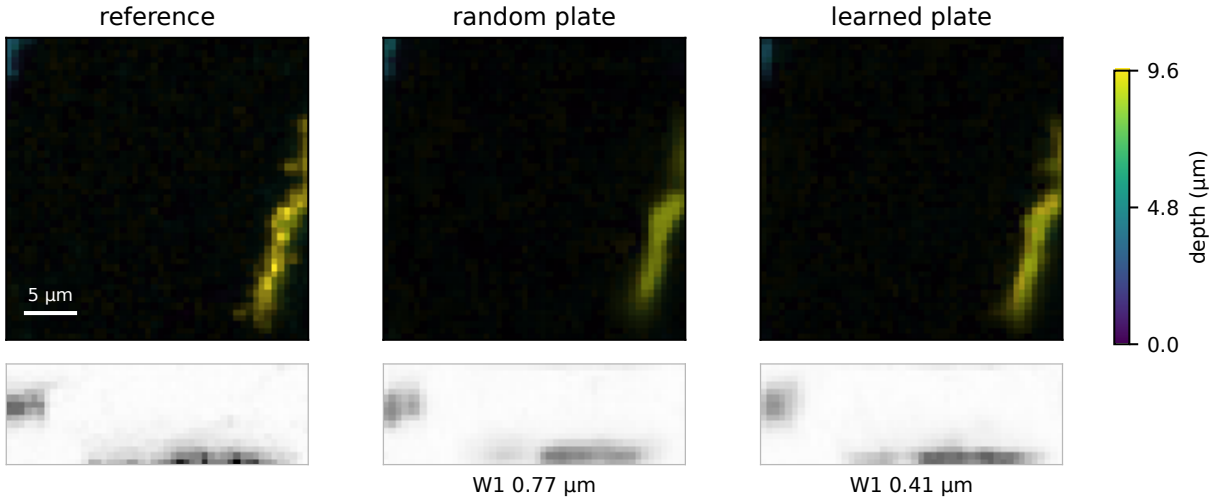


Figure 4. **Neuronal reconstruction with pretrained PnP.** Top: lateral maximum-intensity projections, coloured by the depth of the brightest voxel. Bottom: axial maximum-intensity projections. Each view uses a common display scale across the reference and reconstructions. The crop is selected by median W1 improvement across the three methods; draw 0 is shown at $C = 8$ and equal dose. Selection and all-decoder views are given in the supplement.

reduction does not imply an equal reduction in acquisition time.

At a fixed 32 directions, the learned plate at 0.3 times the excitation dose achieves lower W1 than the random plate at full dose (Fig. 5). Reductions are 26 % [20, 33] with TV, 39 % [34, 45] with pretrained PnP and 22 % [16, 28] with learned PnP. These are paired comparisons at the tested dose levels. The small changes between full and triple dose describe the selected reconstruction protocols, rather than a fundamental photon limit.

6.6. Optical objectives and robustness

We compare depth decorrelation with worst-case coherence, depth Fisher information and an uncentred variant (Tab. 2). All objectives improve upon the random plate. Across both optimisation seeds and all three decoders, our objective gives 13–20 % lower W1 at $C = 8$ than the tested coherence and Fisher-information implementations. Centring has negligible effect. These comparisons favour depth decorrelation under the tested training conditions.

Noise. With background and read noise, W1 reductions remain 30 %, 53 % and 27 % for TV, pretrained PnP and learned PnP. Gains also persist when both encoders receive one tenth of the reference dose. The full noise model and paired results are given in the supplement.

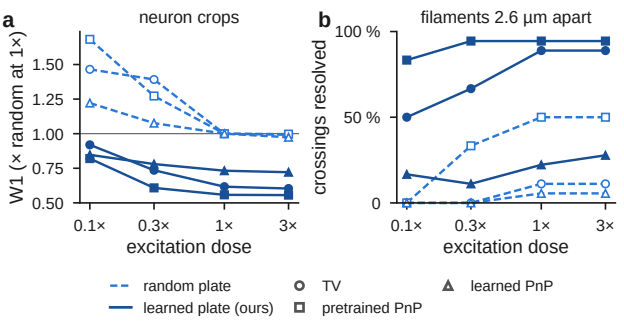


Figure 5. **Benefits at reduced excitation dose.** (a) Neuronal W1 relative to each decoder's random-plate result at full dose, evaluated on the 12 prespecified crops. (b) Success rates for crossing filaments 2.56 μm apart. Solid: learned plate; dashed: random plate. Exposure varies at a fixed count of 32 illumination directions.

	axial W1 (μm) ↓		ours vs this
objective	$C = 1$	$C = 8$	reduction [95 % CI]
random plate	0.203	0.282	27 % [20, 34]
worst-case coherence	0.172	0.238	13 % [12, 15]
depth Fisher information	0.164	0.240	14 % [12, 17]
decorrelation, uncentred	0.142	0.207	0 % [−1, 1]
depth decorrelation (ours)	0.142	0.206	–

Table 2. **Controlled comparison of optical training objectives.** Learned-PnP W1 on the 12 prespecified crops. Plates share the initial phase, sampling and optimisation budget, with validation tuning for each encoder. The final column compares our plate with each row at $C = 8$.

Alignment and calibration. For learned PnP, plate-distance offsets of ± 0.5 μm retain W1 reductions of 27–28 % after updating the forward model. Uncalibrated pointing errors of 0.005–0.01 in direction cosine retain reductions of 13–20 %. In contrast, uncalibrated plate-distance or lateral-position errors eliminate the learned advantage; a one-feature lateral shift makes W1 11 % worse than the random plate (95 % CI: 2–23 %). Accurate optical registration is therefore necessary to preserve the benefit.

7. Discussion and limitations

Learning depth-discriminative illumination. Our results connect an optical design criterion with measurement distinguishability, controlled separation and volumetric recovery. Training uses neither specimen examples nor reconstruction feedback, yet the learned plate improves axial profiles under TV and both PnP configurations. The measurement-level comparison supports an optical contribution before decoding, while the reconstruction results show how that contribution appears under different priors. The two PnP methods share an underlying framework; these findings do not establish transfer to arbitrary decoders or superiority over end-to-end optical learning.

Profile recovery and structural separation. Axial W1 detects differences that a centroid alone can miss, but lower profile error does not imply that every overlapping structure is resolved. The crossing threshold improves under two decoders and remains unchanged under learned PnP. Many layered neuronal columns also remain merged under the evaluated peak criterion. Supplementary set-membership tests find uninformative voxelwise bounds on dense neuronal windows. Improved depth discrimination and reconstruction accuracy therefore do not establish unique neuronal connectivity.

Physical assumptions and implementation. Our results are based entirely on simulations of linear fluorescence in a homogeneous, non-scattering medium with idealised optical components. Scattering, aberrations and other experimental imperfections may alter the depth-dependent illumination patterns and the discrimination they provide. Although the alignment simulations show that the gains can survive moderate perturbations when the optical model is updated, uncalibrated misalignment can eliminate them. Experimental validation requires accurate characterisation of the realised optical system and its measurement operator.

Data and statistical scope. The main comparison uses 60 neuronal crops from six cells across three animals. Crops from the same cell are correlated, and cell-bootstrap intervals remain approximate with only six cells. Objective, dose and tolerance comparisons use the 12 prespecified crops. Both optical optimisation seeds start from the same random plate and differ in training samples; this does not establish independence from the initial phase. The neuron-trained denoisers use a fourth animal. Broader specimen and hardware evaluation is needed to assess the range over which the learned encoding remains effective.

8. Conclusion

We learn a fixed phase plate for depth-encoding excitation through the optical forward model, without specimen examples or a reconstruction decoder. In simulations, the learned measurements better distinguish selected axial alternatives. TV and pretrained PnP separate crossing filaments at a smaller tested gap, while axial-profile W1 falls by 28–50 % across three reconstruction methods at eightfold measurement compression and equal deposited dose. Gains persist at reduced dose. Learning the excitation improves depth discrimination and recovery across the tested reconstruction priors.

LLM Usage Statement

LLMs were used for grammar and fact checking.

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